Supplementary Material: Insider Threat Detection using Machine Learning SLR

Note: this Supplementary material can also be accessed online on GitHub at: https://github.com/alriyami7777-alt/Insider-Threat-ML-SLR-Supplement

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1. Overview

This repository contains the complete supplementary materials and "audit trail" for our Systematic Literature Review (SLR) on Insider Threat Detection using Machine Learning. This review synthesizes **82 primary studies** published between 2021 and 2025.

The repository provides full transparency into our methodology, including our search strategy, the PRISMA-2020 selection process, quality assessment scoring, and the AI-assisted data extraction process and pipeline.

2. Repository Structure

The repository is organized into six folders:

Folder Name	Primary Content	Format
01_Search_Strategy	Boolean search strings, database search logs, and raw record counts. RIS and CSV files from the databases	PDF RIS CSV
02_Study_Selection	PRISMA 2020 flow diagram and the Exclusion Log for the 16 rejected full-text papers.	PDF / CSV
03_Quality_Assessment	Detailed QA1–QA5 scoring	PDF / Excel
04_Data_Extraction	Structured data extraction for RQ1–RQ5, including AI-assisted extraction methodology.	PDF
05_Visualizations	Bibliometric data, JCR quartile distributions, and keyword frequency datasets.	PDF / Excel
06_PRISMA_Checklists	Completed PRISMA 2020 checklist and PRISMA 2020 for Abstract checklist.	PDF

3. Transparency

To ensure the highest level of academic strenght, this repository documents two specialized workflows used in our study:

- **AI-Assisted Extraction (Folder 04):** We utilized **NotebookLM** for the initial extraction of technical parameters from the 82-paper. All AI-generated data were audited and manually verified by the authors.
- **Dual-Stream Bibliometric Pipeline (Folder 05):** Visualizations were generated through a hybrid pipeline combining qualitative thematic scans (NotebookLM) and quantitative mapping (Zotero/Excel).

4. How to Use This Repository

📁 01_Search_Strategy: For Replication & Verification

- **Replicate Searches:** Use the provided Boolean strings to run identical queries across WoS, IEEE Xplore, ScienceDirect, ACM, or Scopus.
- **Audit Raw Data:** The `.ris` and `.csv` files contain the "raw" results before deduplication, allowing you to see exactly what the databases returned on the search date.

📁 02_Study_Selection: For Procedural Transparency

- **Track the Process:** The **PRISMA 2020 Flow Diagram** provides a high-level visual of the inclusion/exclusion phases.
- **Verify Exclusions:** It lists the 16 full-text papers that were rejected and provides the specific technical reason for their removal.

📁 03_Quality_Assessment: For Rigor Validation

- **Review Scoring:** Details on how each of the articles performed against our five Quality Assessment (QA) criteria.

📁 04_Data_Extraction: For Technical Synthesis

- **Answer Research Questions:** The structured data is mapped directly to **RQ1–RQ5**. Use these tables to find specific information on model architectures, datasets used, or feature engineering techniques.
- **AI Methodology:** Review the document on AI-assisted extraction to understand how LLMs were used to accelerate the data synthesis process while maintaining human-in-the-loop accuracy.

📁 05_Visualizations: For Trend Analysis

- **Identify Leading Venues:** Consult the **JCR Quartile distributions** to see which top-tier journals are publishing the most relevant research.
- **Keyword & Bibliometrics:** Use the frequency datasets to identify emerging trends in the literature (e.g., the shift from traditional ML to Deep Learning or Transformer-based models).

📁 06_PRISMA_Checklists: For Standard Compliance

- **Verify Compliance:** Open these PDFs to see exactly where in our main manuscript we addressed each item required by the **PRISMA 2020 reporting standards**.